

Modulus, Argument multiplication

Basic skills

Q1. $z = 9 + 12i$, $w = 3 + i$

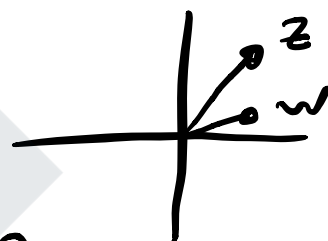
$$x + yi$$
$$\sqrt{x^2 + y^2}$$
$$\tan^{-1}\left(\frac{y}{x}\right)$$

a) $|z| = \sqrt{9^2 + 12^2} = 15$

$$\text{Arg } z = \tan^{-1}\left(\frac{12}{9}\right) = 0.927$$

$$|w| = \sqrt{3^2 + 1^2} = \sqrt{10}$$

$$\text{Arg } w = \tan^{-1}\left(\frac{1}{3}\right) = 0.322$$



b) $zw = (9 + 12i)(3 + i)$

$$= 27 + 9i + 36i + 12i^2$$

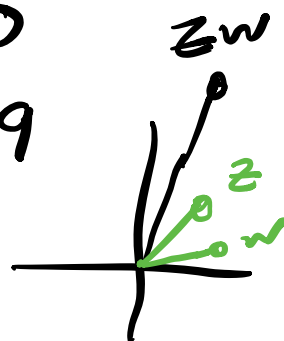
$$= 27 + 45i - 12$$

$$= 15 + 45i$$

$$i^2 = -1$$

c) $|zw| = \sqrt{15^2 + 45^2} = 15\sqrt{10}$

$$\text{Arg } zw = \tan^{-1}\left(\frac{45}{15}\right) = 1.249$$



$$d) |zw| = 15\sqrt{10} = |z||w|$$

$$\text{Arg } zw = 1.249 = \text{Arg } z + \text{Arg } w$$

Competency skills

Q2. $zw = 2 \times 5 \left(\cos\left(\frac{\pi}{4} + -\frac{\pi}{3}\right) + i \sin\left(\frac{\pi}{4} + -\frac{\pi}{3}\right) \right)$

multiply modulus $= 10 \left(\cos\left(-\frac{\pi}{12}\right) + i \sin\left(-\frac{\pi}{12}\right) \right)$

Add argument

Q3. $zw = 2\sqrt{3} \times \sqrt{3} \left(\cos\left(\frac{\pi}{6} + \frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{6} + \frac{\pi}{2}\right) \right)$

$$= 6 \left(\cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right)$$

Q4. $z = x + yi$, $w = u + vi$

$$zw = (x + yi)(u + vi)$$

$$= xu + xvi + yui + yvi^2$$

$$= (xu - yv) + (xv + yu)i$$

$$|z| = \sqrt{x^2 + y^2} \quad |w| = \sqrt{u^2 + v^2}$$

$$|zw| = \sqrt{(xu - yv)^2 + (xv + yu)^2}$$

$$= \sqrt{(x^2 + y^2)(u^2 + v^2)}$$

$$= |z| |w|$$

$$\text{Q5. a) } z = \frac{\sqrt{2}}{2} (1+i) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i$$

$$|z| = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{1} = 1$$

$$\text{Arg } z = \tan^{-1}\left(\frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}}\right) = \tan^{-1} 1 = \frac{\pi}{4}$$

$$z = \cos\left(\frac{\pi}{4}\right) + i \sin\left(\frac{\pi}{4}\right)$$

Special values of cos and sin

$$\text{b) } 2z \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$

arg of other complex number

mod of other complex number

$$= 2 \times 1 \left(\cos\left(-\frac{\pi}{4} + \frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4} + \frac{\pi}{4}\right) \right)$$

$$= 2 (\cos 0 + i \sin 0)$$

$$= 2$$

Q6. a)

$$\frac{2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}{5 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)}$$

$$= \frac{2}{5} \left(\cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \right)$$

$$= \frac{2}{5} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

b) $\frac{\sqrt{3} \cos \left(-\frac{2\pi}{3} \right) + \sqrt{3} i \sin \left(-\frac{2\pi}{3} \right)}{\sqrt{12} \cos \left(\frac{2\pi}{3} \right) + \sqrt{12} i \sin \left(\frac{2\pi}{3} \right)}$

$$= \frac{\sqrt{3}}{\sqrt{12}} \left(\cos \left(-\frac{2\pi}{3} - \frac{2\pi}{3} \right) + i \sin \left(-\frac{2\pi}{3} - \frac{2\pi}{3} \right) \right)$$

$$= \sqrt{\frac{1}{4}} \left(\cos \left(-\frac{4\pi}{3} \right) + i \sin \left(-\frac{4\pi}{3} \right) \right)$$

$$= \frac{1}{2} \left(\cos \left(\frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} \right) \right)$$

Notice

$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|}$$

$$\text{Arg} \left(\frac{z}{w} \right) = \text{Arg} z - \text{Arg} w$$

are opposites of multiply

$$c) \frac{\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right)}{2i}$$

$$= \frac{\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right)}{2 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)}$$

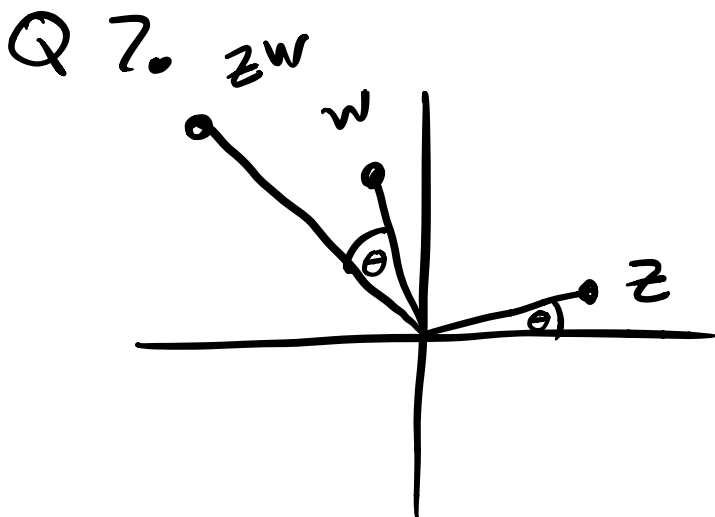
convert
form

$$= \frac{1}{2} \left(\cos\left(-\frac{2\pi}{3} - \frac{\pi}{2}\right) + i \sin\left(-\frac{2\pi}{3} - \frac{\pi}{2}\right) \right)$$

$$= \frac{1}{2} \left(\cos\left(-\frac{7\pi}{6}\right) + i \sin\left(-\frac{7\pi}{6}\right) \right)$$

$$= \frac{1}{2} \left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right) \right)$$

Advanced skills



The multiplication transforms w by z i.e. it enlarges it by $|z|$ and rotates by $\theta = \text{Arg } z$

Q8. $z = 2\sqrt{3}i - 2$

a) $|z| = \sqrt{(2\sqrt{3})^2 + (-2)^2} = \sqrt{16} = 4$

$\text{Arg } z = \tan^{-1}\left(\frac{2\sqrt{3}}{-2}\right) = -\frac{\pi}{3}$
 so $\frac{2\pi}{3}$

$z = 4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$

b) $z^2 = 16 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right)$
 $= 16 \left(\cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \right)$

$z^3 = 64 \left(\cos 0 + i \sin 0 \right) = 64$

$z^4 = 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$

$z^5 = 1024 \left(\cos -\frac{2\pi}{3} + i \sin -\frac{2\pi}{3} \right)$

c) $z^{3n} = (z^3)^n = 64^n$

d) $z = 2\sqrt{3}i - 2 \leftarrow \text{as } z^3 = 64$